National Steel Swaging Sleeves

S-505 COLD TUFF® Standard Steel Sleeves

S-505 Standard Steel Sleeve Specifications												Swager / Die Data		
	Rope Size				Before Swage Dimensions (in.)					Maximum After Swage Dimensions (in.)		Standard Round Dies		QUIC-PASS Dies
S-505 Stock No.	(in.)	(mm)*	Weight Per 100 (lbs.)	Pkg. Qty.	A	B	D	E	G	Standard Die	QUIC-PASS Die †	Die Description	Standard Die Stock No.	QUIC-PASS Die Stock No. †
1041063	1/4	6-7	5	250	1.00	.66	.31	.28	.47	.57	.565	1/4 Taper	1197528	1923530
1041090	5/16	8	14	200	1.50	.91	.44	.44	.62	.75	.769	3/8 Taper	1192364	1923551
1041107	3/8	9-10	14	100	1.50	.91	.47	.39	.66	.75	.769	3/8 Taper	1192364	1923551
1041125	7/16	11	33	50	2.00	1.22	.55	.65	.85	1.01	1.016	1/2 Taper	1192408	1923572
1041143	1/2	13	29	50	2.00	1.22	.63	.56	.91	1.01	1.016	1/2 Taper	1192408	1923572
1041161	9/16	14	64	25	2.75	1.47	.69	.63	1.03	1.24	1.247	5/8 Taper	1192444	1923593
1041189	5/8	16	56	25	2.75	1.47	.75	.63	1.09	1.24	1.247	5/8 Taper	1192444	1923593
1041205	3/4	18-19	88	20	3.19	1.72	.91	.84	1.28	1.46	1.475	3/4 Taper	1192462	1923614
1041223	7/8	22	131	10	3.56	2.03	1.03	1.00	1.53	1.68	1.738	7/8 Taper	1192480	1923635
1041241	1	25-26	195	10	4.00	2.28	1.16	1.13	1.72	1.93	1.955	1 Taper	1192505	1923656
1041269	1-1/8	28-29	260	Bulk	4.81	2.50	1.28	1.25	1.94	2.13	2.170	1-1/8 Open 1st Stage 2nd Stage	 1192523 1192541	1923677
1041287	1-1/4	31-32	355	Bulk	5.19	2.78	1.44	1.41	2.16	2.32	2.405	1-1/4 Open 1st Stage 2nd Stage	 1192621 1192587	1923698
1041303	1-3/8	34-35	423	Bulk	5.81	3.00	1.56	1.56	2.38	2.52	2.610	1-3/8 Open 1st Stage 2nd Stage	 1192667 1192621	1923717
1041321	1-1/2	37-38	499	Bulk	6.25	3.25	1.69	1.69	2.63	2.71	2.835	1-1/2 Open 1st Stage 2nd Stage	 1192649 1192667	1923736

* For metric rope sizes not shown, see Intermediate Metric Die Chart on page 25.
† QUIC-PASS® system not available for 12mm, 20mm, 24mm and 36mm wire rope sizes.

S-505 Standard Steel Sleeve Specifications											Swager / Die Data					
	Rope Size				Before Swage Dimensions (in.)					Maximum After Swage Dimensions (in.)		Stock No.				
												500 Tons 1000 Tons 1500 Tons 5x7	Front Load		Side Load	
S-505 Stock No.	(in.)	(mm)	Weight Per 100 (lbs.)	Pkg. Qty.	A	B	D	E	G		Die Description		1500 Ton 6x12	3000 Ton 6x12	1500 Ton 6x12	3000 Ton 6x12
1041349	1-3/4	44-45	805	Bulk	7.25	3.84	1.94	1.97	3.13	3.10	1-3/4 Open 1st Stage 2nd Stage	1192685 1192701	—	—	—	—
1041367	2	50-52	1132	Bulk	8.50	4.38	2.25	2.25	3.63	3.56	2 Open 1st Stage 2nd Stage	1192729 1192747	—	—	—	—
1041385	2-1/4	56-57	1936	Bulk	9.56	5.03	2.50	2.53	4.03	4.12	2-1/4 Open 1st Stage 2nd Stage	1192765 1192783	1191089 1191043	1191089 1191043	—	1195085 1195067
1041401	2-1/2	62-64	2352	Bulk	10.50	5.50	2.75	2.81	4.50	4.50	2-1/2 Open 1st Stage 2nd Stage	—	1191061 1191089	1191061 1191089	1195370 1195469	1195076 1195085
1041429	2-3/4	68-70	2800	Bulk	11.50	5.75	3.00	3.09	4.75	4.70	2-3/4 Open 1st Stage 2nd Stage	—	1191034 1191052	1191034 1191052	1195389 1195478	1195094 1195101
1041447	3	75-76	2940	Bulk	12.00	6.00	3.25	3.38	5.00	4.96	3 Open 1st Stage 2nd Stage	—	1193201 1193229	1193201 1193229	1195398 1195487	1195110 1195129
1041483	3-1/2	87-89	4640	Bulk	14.00	7.00	3.88	3.94	5.84	5.77	3-1/2 Open 1st Stage 2nd Stage	—	1193247 1193265	1193247 1193265	—	1195138 1195147
1041492	3-3/4	93-95	5500	Bulk	15.00	7.50	4.06	4.25	6.31	6.23	3-3/4 Open 1st Stage 2nd Stage	—	—	1191114 1191132	—	1195263 1195272
1041508	4	100-105	6800	Bulk	16.00	8.13	4.38	4.50	6.81	6.69	4 Open 1st Stage 2nd Stage	—	—	1191150 1191178	—	1195156 1195165
1041526	4-1/2	112-114	10000	Bulk	18.00	9.13	4.88	5.06	7.66	7.45	4-1/2 Open 1st Stage 2nd Stage	—	—	1191187 1191203	—	1195174 1195183

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Intermediate Metric Die Chart

Sleeve and Swaging Die Requirements for Intermediate Sizes of Metric Wire Rope							
S-505 Stock No.	S-505 Sleeve Size	Metric Wire Rope Size	Standard Round Dies				Maximum After Swage Dimension
			1st Stage Die		2nd Stage Die		
1041143	1/2	12	1190881	5 x 7 Double Cavity	—		.990
1041223	7/8	20	1190901	5 x 7 Double Cavity	—		1.620
1041241	1	24	1190921	5 x 7 Double Cavity	—		1.880
1041321	1-1/2	36	1192649	5 x 7	1190941	5 x 7	2.630
1041349	1-3/4	40	1192685	5 x 7	1190961	5 x 7	2.950
1041367	2	48	1192729	5 x 7	1190971	5 x 7	3.460
1041401	2-1/2	60	1192809	5 x 7	1190981	5 x 7	4.370
1041401	2-1/2	60	1191061	6 x 12	1190991	6 x 12	4.370
1041487	3	72	1193201	6 x 12	1191001	6 x 12	4.810
1041483	3-1/2	80	1193247	6 x 12	1191101	6 x 12	5.450
1041483	3-1/2	84	1193247	6 x 12	1191121	6 x 12	5.550

QUIC-PASS® system not available for these metric rope sizes.

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Flemish Type Wire Rope Sling Eyes – Introduction

Flemish type wire rope sling eyes are formed by looping an eye (see information on pages 27-28 for forming eye properly) and cold swaging a steel sleeve on the rope at the base of the loop. Swaging bonds the ends of the rope strands in place by forming a streamlined, compact, solid and permanent splice. The Flemish wire rope sling eyes have all the advantages of a hand spliced eye with the added safety of a swaged steel sleeve to complete the termination while leaving no rope distortion, wire ends or chance of rope slippage due to rotation.

The Flemish Eye ensures equal distribution of the load in both parts of the eye because pull is directly along the center line of the rope and eye. Cold swaging the National S-505 Standard Steel Sleeve forms a solid bond around the splice which will achieve a breaking strength near the rated breaking strength of the wire rope, thus giving maximum efficiency of the termination (see page 23 for rated efficiencies).

The National S-505 COLD TUFF® Steel Sleeves are engineered to be cold swaged without failure (when proper swaging techniques as described on pages 27-35 are used) in both extreme hot or cold environments. One end of the sleeve is tapered to allow the splice to pass under obstructions without snagging. The other end has a radius which helps to eliminate rope abrading.



Swaged Flemish Eye Termination using a National S-505 "COLD TUFF®" Standard Steel Sleeve.

Forming the Eye

In accordance with the Wire Rope Technical Board's Sling committee's recommendation, we endorse that a seven (7) lay rope eye will give maximum efficiency with no advantage in using more.

Separating Rope Strands

Flemish Eyes are formed by separating the strand into two groups with the core and 3 strands in one group and 3 strands in the other group. (See page 28, Step 1.)

The stopping point for separating the strands determines the length of the strand ends at the throat of the splice. If the strand ends or pull amount are too short, relay the strands and insert one strand to the left. If the strand ends or pull are too long, move one strand to the right. See pages 27-28 for detail steps in marking and making a Flemish Eye.

Important

The Flemish Eye for the QUIC-PASS® system is formed in the same manner as the standard round sleeves. See page 35 for detail steps in marking a QUIC-PASS® Flemish Eye.

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Flemish Type Wire Rope Sling Eyes – Marking

Marking & Calculations for Forming Mechanical Spliced Flemish Style Eyes

Calculation for Forming Different Size Eyes

Many methods have been used to form the Flemish Style Eye (i.e., from counting the number of lays to unlay to trial and error). The method used here is based on a mathematical formula that will produce the required results on the rope recommended by Crosby National Swage.

Standard Eyes (7 lay)
Pull amount or tail is equal to 3.75 X diameter (d) of rope. Throat amount is equal to 40 X diameter (d) of rope. See example below.

Standard Thimble Eyes
Pull amount is equal to 3.75 X diameter (d) of rope. Throat amount is equal to 15 X diameter (d) of rope. See example below (change throat amount to 15).

Larger than Standard Eyes
Pull amount or tail is equal to 3.75 X diameter (d) of rope. Throat amount is equal to 5 X width of eye to be formed. Note: All standard eyes have a width of ½ the length (Example: 6" wide X 12" long eye).

Use assembly procedure for Flemish Eye on Page 28.

Example: Standard Eye (7 lay) for 1/2" wire rope

Step 1: To find the Pull Amount or tail, Mark 1 (See Figure 1).

Mark 1 = 3.75 X d (d is the diameter of the rope)

Example: Mark 1 = 3.75 X .5" = 1.875"
(Diameter of Rope)

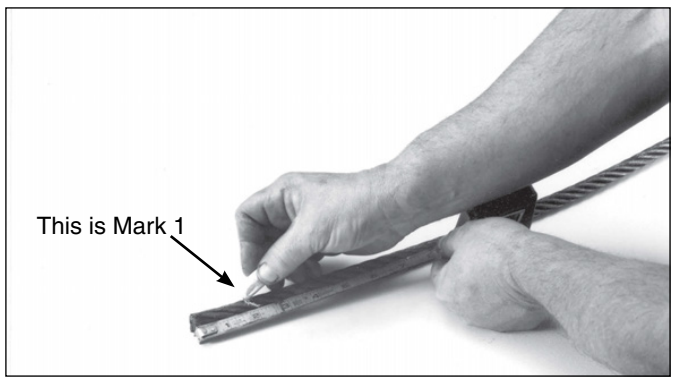


Figure 1

Step 2: To find the throat, Mark 2 (See Figure 2).

Mark 2 = 40.00 X d

Example: Mark 2 = 40.00 X.5 = 20.00
(Standard Throat Constant) (Diameter of Rope)

Measure from Mark 1

Total amount of rope to form Standard eye including tail (from end of rope) for Flemish style splice is 21.875".

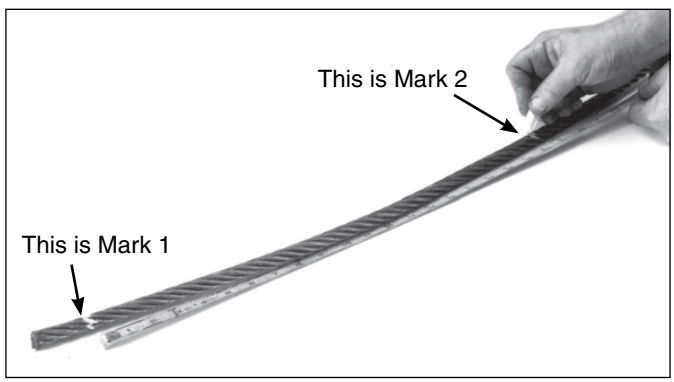


Figure 2

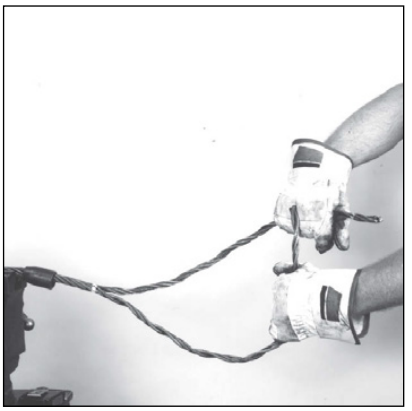
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Flemish Type Wire Rope Sling Eyes

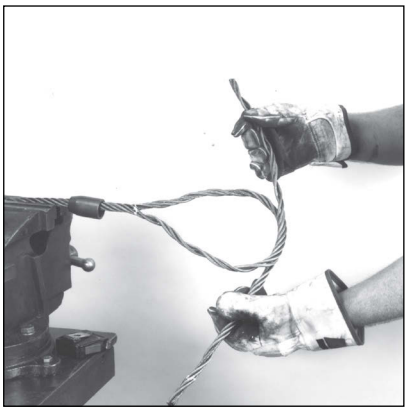
Fabrication Procedure for Mechanical Spliced Flemish Eyes

We recognize that there are other ways to produce a Flemish Eye. However, this is one way that can be employed.

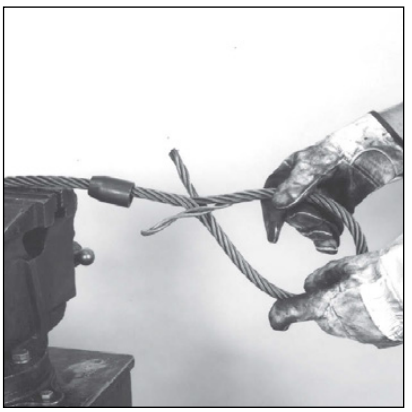
Step 1: Transfer your marks 1 and 2 to your wire rope (See page 27 for instructions). Slide proper sleeve on rope and unlay the rope dividing it into 2 sections consisting of 3 strands and 3 strands plus the core. See page 23 for recommended rope to use.



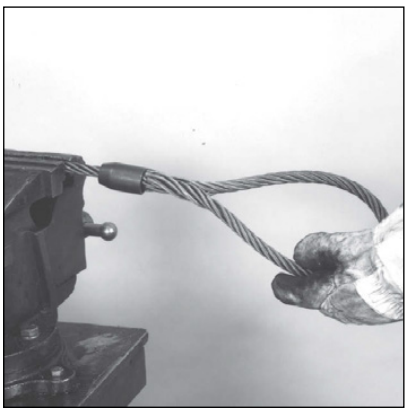
Step 2: Cross Mark 1 of the 3 strand group over the core group at Mark 2. Then bring the core group over the 3 strand group until they lock together in their natural configuration. Keep Marks 1 and 2 as close together as possible.



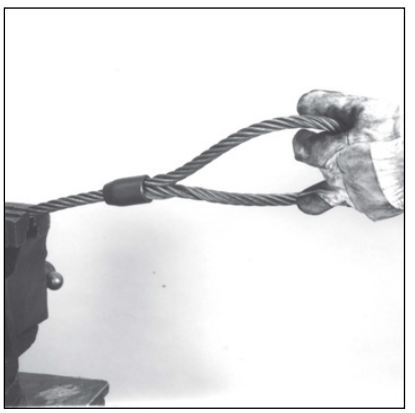
Step 3: Relay or cross over the core and strand group until eye is formed.



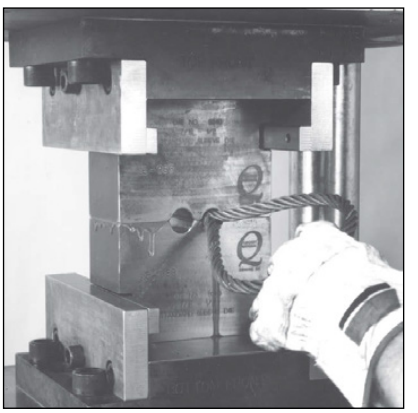
Step 4: Relay the rope completely, including tails, or pull amount of rope. Our tests have proven that when the tail's ends, or pull amount of rope, are on the outside of the rope you achieve the greatest efficiency. **Making Flemish Eyes by tucking the tail's ends, or pull amount, on the inside of the rope is not recommended (The Texas Tuck).**



Step 5: Slide sleeve over tail and make sure the sleeve is all the way upon tail before swaging. This can be easily achieved by using a flat bar or a hammer and driving the sleeve, from the nose end, until it is fully seated over the tail. The rope at the end of the sleeve should then be marked. This will allow you to determine if the sleeve has moved before and during swaging.



Step 6: Swage product using proper swaging methods on pages 29-35. Swage with proper dies to ensure required after swage dimension (see pages 24-25 in this manual, or look up proper die and maximum after swage dimensions in either The Crosby General Catalog, the National Swage Die Guide, or the National Swage Die Chart). Check the fitting with proper measuring devices or No-Go gauges to insure the correct swage dimensions. See pages 17-18 for proper way to check fittings after swage dimensions.



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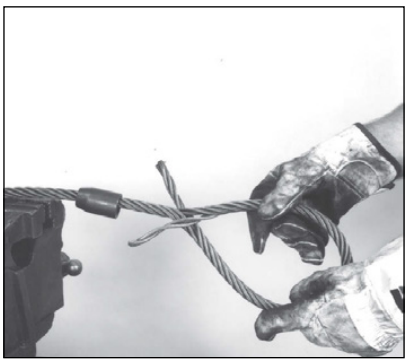
Flemish Type Wire Rope Sling Eyes – ¼" - 1"

Flemish Eye Terminations for 1/4" thru 1" National S-505 Sleeves. For larger sizes, see pages 31-34. Dies for National S-505 Standard Steel Sleeve, sizes 1/4" thru 1", are provided with two cavities. **National 1/4" through 1" S-505 sleeves are designed to be swaged only in the tapered cavity of a two cavity die in a National Hydraulic Swaging Machine.**

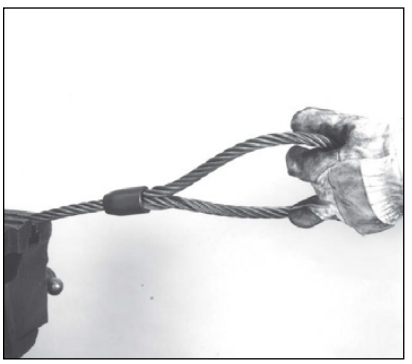
Step 1. Slide sleeve over rope.



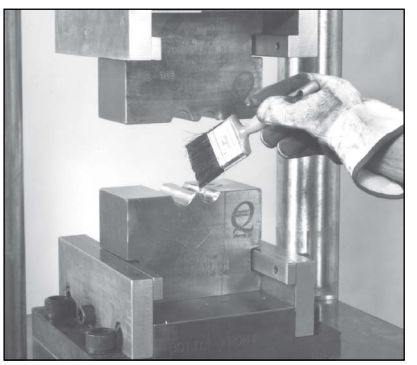
Step 2. Assemble Flemish Eye. See page 28.



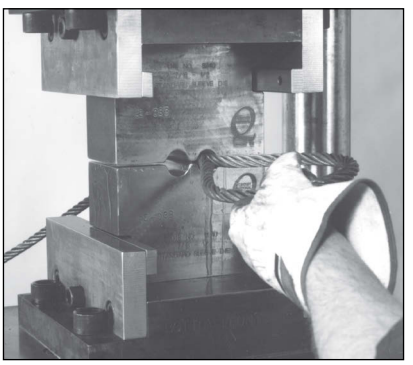
Step 3. Slide sleeve fully over ends of wire rope.



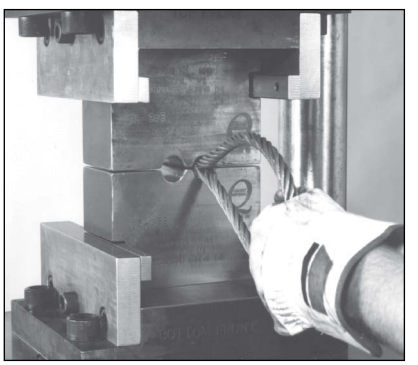
Step 4. Lubricate both die cavities and lower die face. See page 24 for proper die selection.



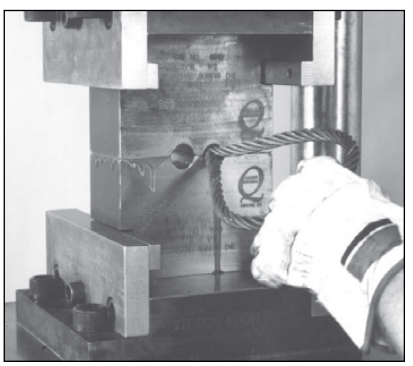
Step 5. 1st pass – allow dies to remain open approximately ½ the distance from time initial contact is made between sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 6. 2nd pass – allow dies to remain open approximately ½ the distance from time initial contact is made between sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 7. 3rd pass – If sharp flashing does not occur, close die. Open dies and rotate fitting 45 to 90 degrees.



Step 8. 4th pass -— Close dies to round fitting. Open dies and rotate fitting 45 to 90 degrees between passes. **Open dies and check for proper after swage dimensions** (See page 17 for proper use of No-Go Gauges).



Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed in tapered die cavity before swaging is complete.

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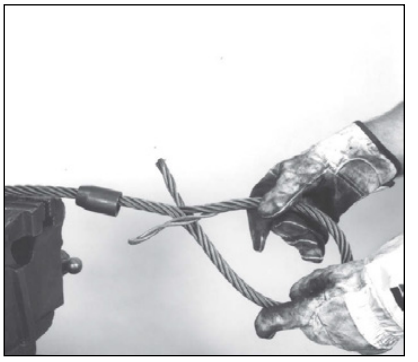
Flemish Type Wire Rope Sling Eyes – 12mm, 20mm & 24mm

Flemish Eye Terminations for 1/2 – 1" National S-505 sleeves on 12mm, 20mm and 24mm metric ropes. For larger sizes see page 31-34. Special dies are required to swage National S-505 sleeves on 12mm, 20mm and 24mm wire ropes. The dies are provided with two cavities. **National 1/2" through 1" S-505 sleeves require swaging in the open cavity and then completing the swaging in the tapered cavity in a National Hydraulic Swaging Machines.**

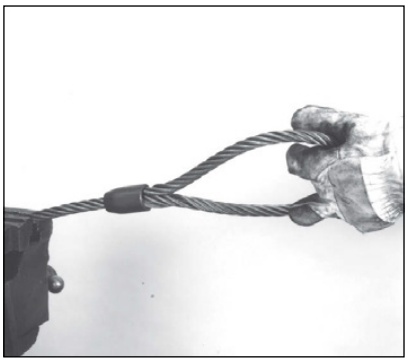
Step 1. Slide sleeve over rope.



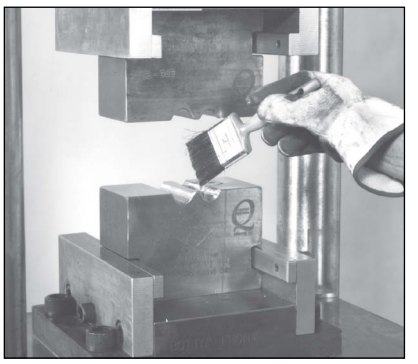
Step 2. Assemble Flemish Eye. See page 28.



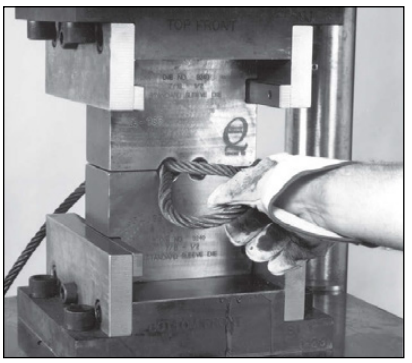
Step 3. Slide sleeve fully over ends of wire rope.



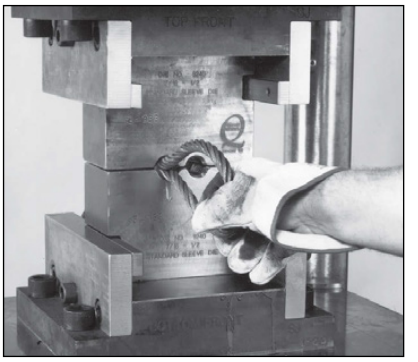
Step 4. Lubricate both die cavities and lower die face. See page 24 for proper die selection.



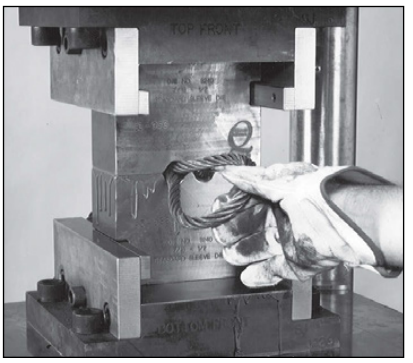
Step 5. 1st pass – In open cavity, allow dies to remain open approximately ½ the distance from time initial contact is made between sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 6. 2nd pass – allow dies to remain open approximately ½ the distance from time initial contact is made between sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 7. 3rd through 6th pass – If sharp flashing does not occur, close dies. **Open dies. Move assembly to tapered cavity and repeat steps 5, 6 and 7.**



Step 8. 7th pass -– Close dies to round fitting. Open dies and rotate fitting 45 to 90 degrees between passes. **Open dies and check for proper after swage dimensions** (See page 17 for proper use of No-Go Gauges).



Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed in tapered die cavity before swaging is complete.

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Flemish Type Wire Rope Sling Eyes – 1⅛" - 1¾"

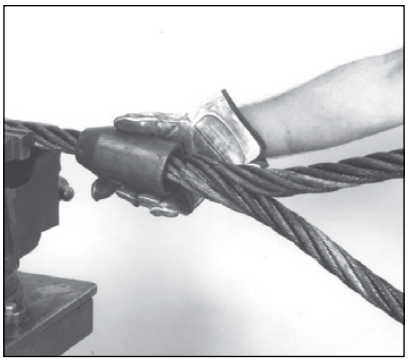
Flemish Eye Termination for 1-1/8" through 1-3/4" S-505 Sleeves. For other sizes, see pages 29-30 and 32-33. National S-505 Standard Steel Sleeve dies are provided with a 1st stage and 2nd stage die. **National S-505 Standard Steel Sleeves 1-1/8" and larger require swaging in the 1st stage die and completing the swaging in 2nd stage dies in a National Hydraulic Swaging Machine.**

1st STAGE

Step 1. Slide sleeve over rope.



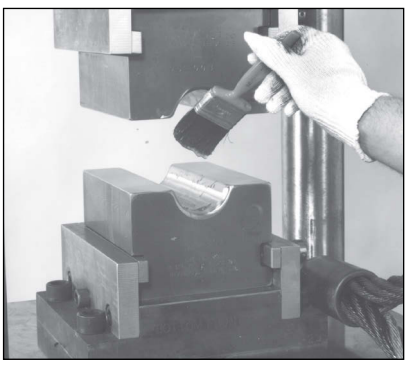
Step 2. Assemble Flemish Eye. See page 28 for assembly instructions.



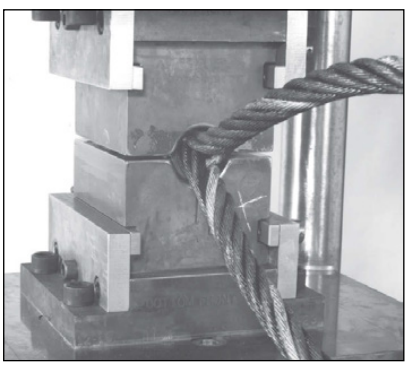
Step 3. Slide sleeve fully over ends of wire rope.



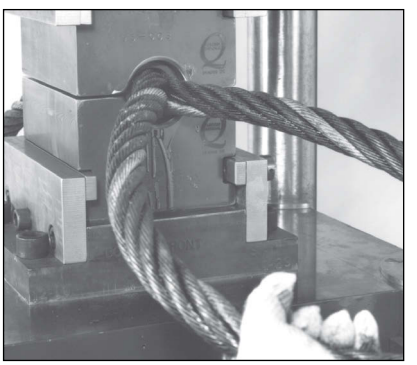
Step 4. Lubricate both die cavities and lower die face. See page 24 for proper die selection.



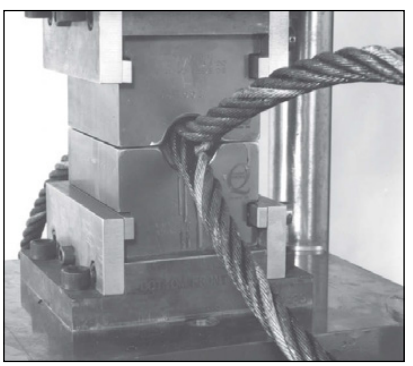
Step 5. 1st pass – allow dies to remain open approximately ½ the distance from time initial contact is made between sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



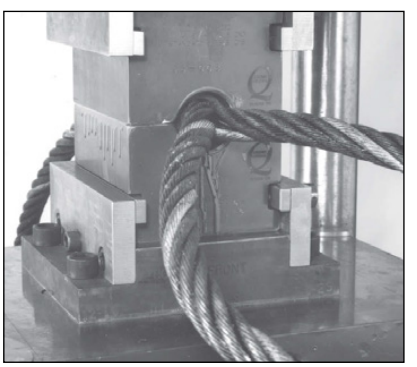
Step 6. 2nd pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 7. 3rd pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 8. 4th and 5th pass – If sharp flashing does not occur, close die. Open dies and rotate fitting 45 to 90 degrees between each pass.



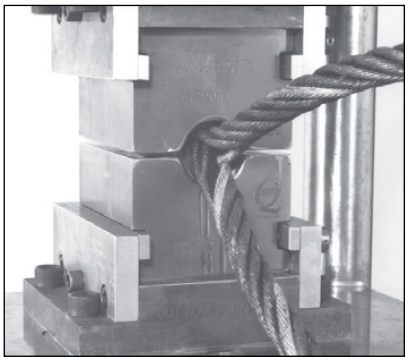
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Flemish Type Wire Rope Sling Eyes – 1⅛" - 1¾"

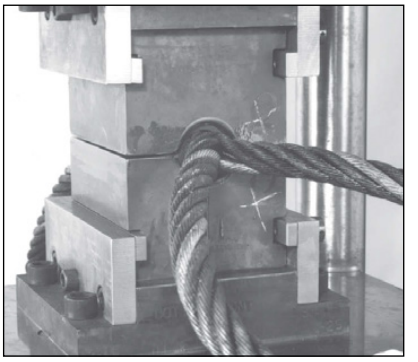
Swaging Instruction for Flemish Eye 1 ⅛"- 1¾" Continued

2nd Stage – Lubricate both die cavities and lower die face. **See page 24 for proper die selection.**

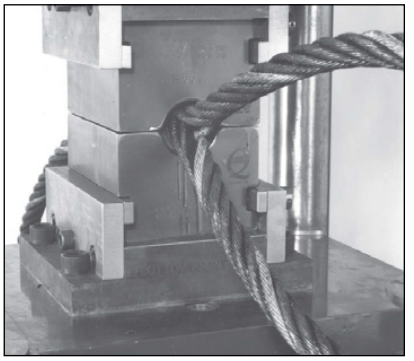
Step 9. 1st pass – allow dies to remain open approximately ½ the distance from time initial contact is made between sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



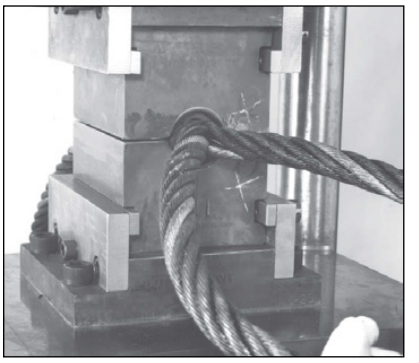
Step 10. 2nd pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.

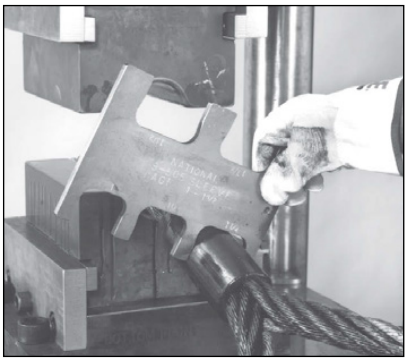


Step 11. 3rd pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 12. 4th pass – If sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees between each pass.





Step 13. 5th and 6th pass – Close dies to round fitting. Open dies and rotate fitting 45 to 90 degrees between each pass. **Open dies and check termination for proper after swaging dimensions.** (See page 17 for proper use of No-Go Gauges).

Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed in 2nd stage die cavity before swaging is complete.

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Flemish Type Wire Rope Sling Eyes – 2″ & Larger

Flemish Eye Terminations for 2” and larger National S-505 Sleeves. For other sizes see pages 29-32. National Standard Steel Sleeve dies are provided with a 1st stage die and 2nd stage dies. **National S-505 Standard Steel Sleeves 1-1/8” and larger require swaging in the 1st stage dies and completing the swage in 2nd stage dies in a National Hydraulic Swaging Machine.**

1st STAGE

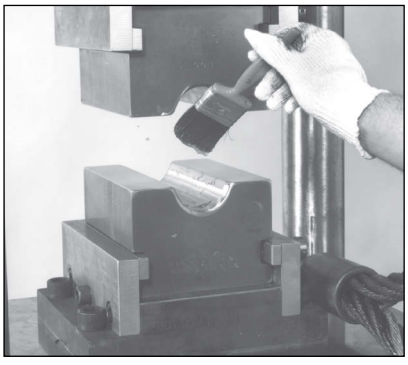
Step 1. Slide Sleeve over rope. Assemble Flemish Eye. See page 28 for assembly instructions.



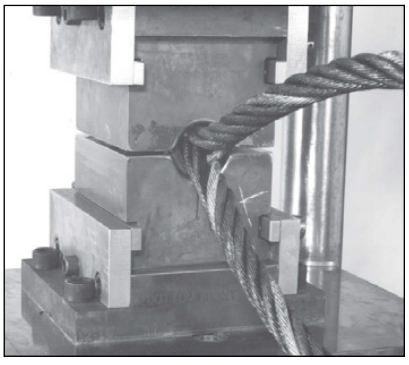
Step 2. Slide sleeve fully over ends of wire rope.



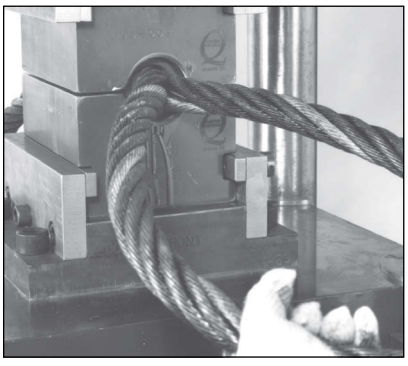
Step 3. Lubricate both die cavities and lower die face. See page 24 for proper die selection.



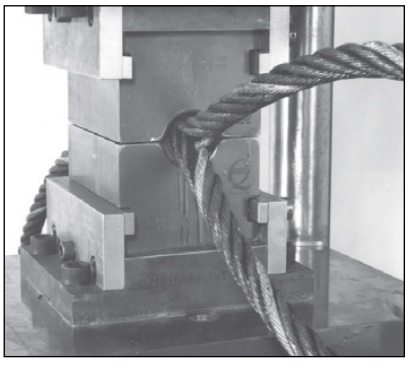
Step 4. 1st pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between sleeve and dies. Rotate fitting 45 to 90 degrees.



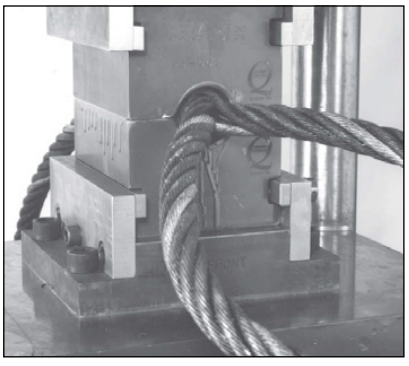
Step 5. 2nd pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between sleeve and dies. Rotate fitting 45 to 90 degrees.



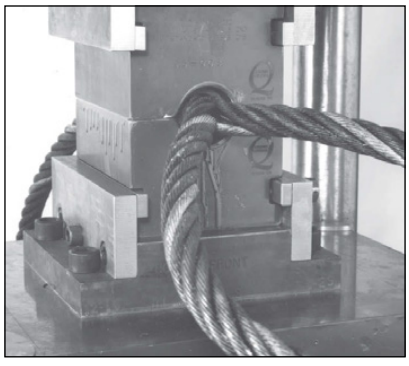
Step 6. 3rd pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between sleeve and dies. Rotate fitting 45 to 90 degrees.



Step 7. 4th pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between sleeve and dies. Rotate fitting 45 to 90 degrees.



Step 8. 5th and 6th pass – If sharp flashing does not occur, close die. Open dies and rotate fitting 45 to 90 degrees between passes.

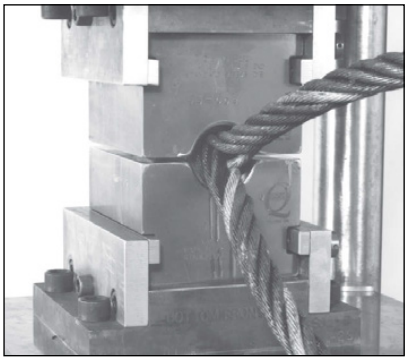


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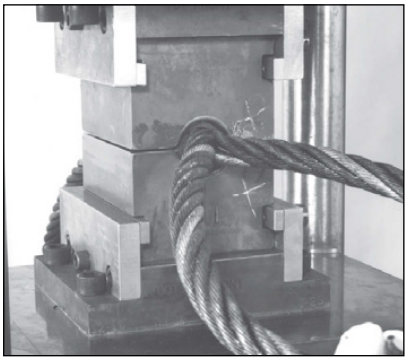
Flemish Type Wire Rope Sling Eyes – 2″ & Larger

2nd Stage – Lubricate both die cavities and bottom die face. **See page 24 for die selection.**

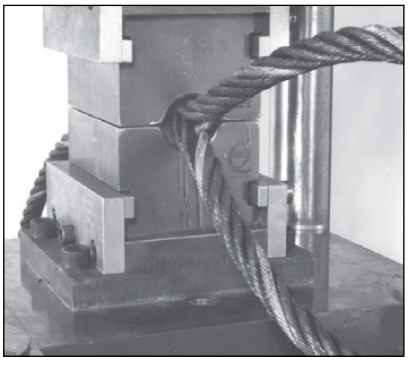
Step 9. 1st pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Rotate fitting 45 to 90 degrees.



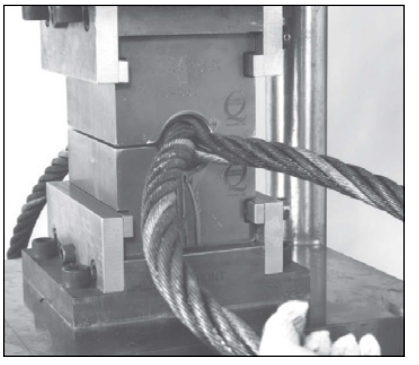
Step 10. 2nd pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Rotate fitting 45 to 90 degrees.



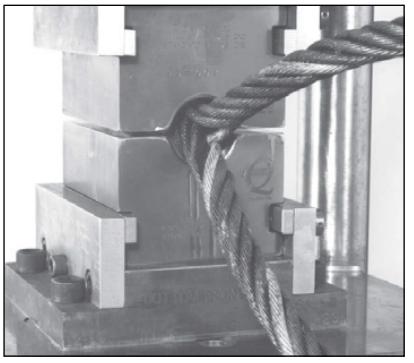
Step 11. 3rd pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the sleeve and dies. Rotate fitting 45 to 90 degrees.



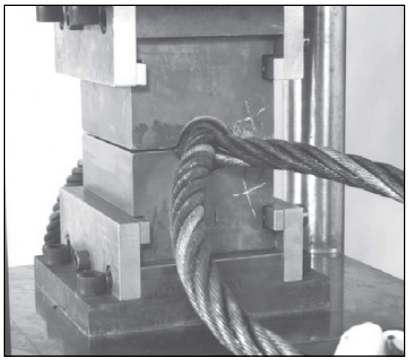
Step 12. 4th pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the sleeve and dies. Rotate fitting 45 to 90 degrees.



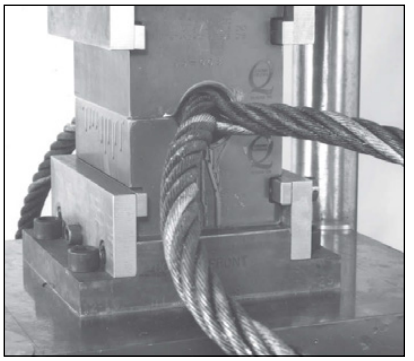
Step 13. 5th pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Rotate fitting 45 to 90 degrees.



Step 14. 6th pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the sleeve and dies. Rotate fitting 45 to 90 degrees.



Step 15. 7th pass – If sharp flashing does not occur, close dies. Rotate fitting 45 to 90 degrees.



Step 16. 8th and 9th pass – Close dies to round fitting. Open dies and rotate fitting 45 to 90 degrees between passes. **Open die and check for proper after swage dimensions.** (See page 17 for proper use of No-Go Gauges).



Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed in 2nd stage die before swaging is complete.

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QUIC-PASS® Swaging Procedures for Flemish Eye Terminations

QUIC-PASS® Swaging Procedures

Flemish Eye Terminations for 1/4" thru 1-1/2" S-505 Sleeves using the QUIC-PASS® Swaging System

QUIC-PASS® dies for National S-505 Standard Steel Sleeves 1/4" thru 1-1/2" are provided with one cavity. National S-505 1/4" thru 1-1/2" sleeves are designed to be swaged in either National standard dies or National QUIC-PASS® dies.

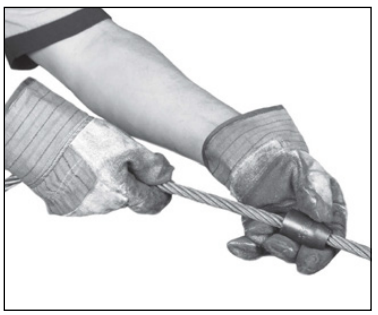
The QUIC-PASS® Swaging System for a National 500 Ton Swaging Machine includes the QUIC-PASS® dies, QUIC-PASS® die holders, a special guide post system and QUIC-PASS® No-Go Gauges. The QUIC-PASS® die holder will also work with National 5 x 7 standard swaging dies. The special guide post system is not required on National Swaging machines larger than 500 Tons. QUIC-PASS® dies for other makes of swaging machines are available (QUIC-PASS® Swaging procedures for other makes of swaging machines are the same as shown below). Die lubrication is critical for the optimum life of the Quic-Pass dies. Lubrication must be applied to the sleeve or die and maintained on the contacting surfaces for each pass during the swage process. The recommended lubrication contains special EP lubricity additives, and is used for extreme drawing operations. Approved lubrication: Shell Fenella Fluid D 2628 at full concentration, or equivalent. Per TDS REF Instructions Doc No. 040629.

Use only National S-505 Standard Steel Sleeves

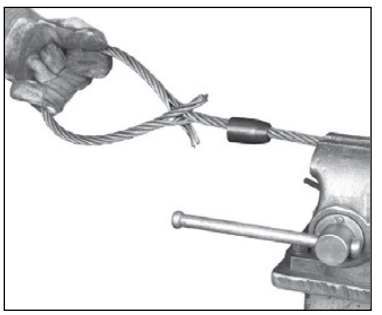
National S-505 Standard Steel Sleeves, when used with the QUIC-PASS® Swaging System, are only recommended for use with one (1) part 6 X 19 or 6 X 36, IPS or XIP (EIP), XXIP (EEIP), RRL, IWRC wire rope.

Before using any National Swage QUIC-PASS® Swaging System with any other type lay, construction, or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

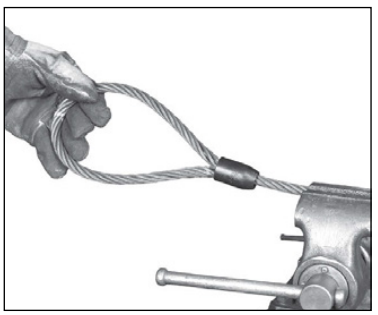
Step 1. Slide S-505 Standard Steel Sleeve over the wire rope.



Step 2. Construct Flemish Eye. See page 28 for assembly instructions.

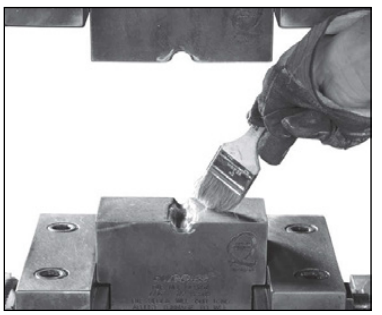


Step 3. Slide S-505 sleeve fully over ends of wire rope to total sleeve depth.



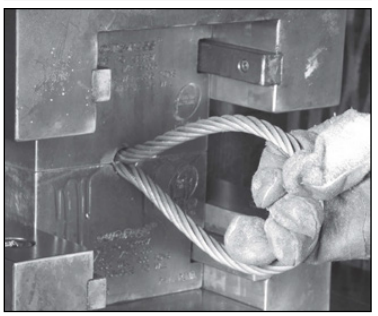
Step 4. Oil lubricate both cavities (top and bottom) and lower die face as well as the sleeve. See page 24 for proper die selection.

Oil lubrication is critical in order to obtain proper after swage dimensions and performance.



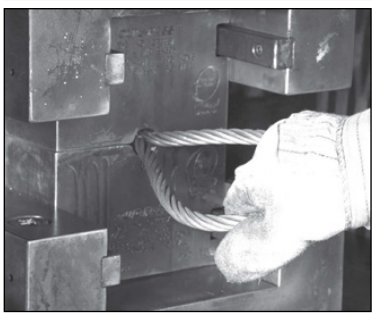
Step 5. 1st pass – Place sling eye in the vertical plane, and close dies completely. Open dies and rotate fitting one flat (60°).

Oozing of the oil from between the die faces will indicate when dies are fully closed.



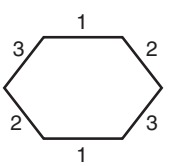
Step 6. 2nd pass – Close dies completely.

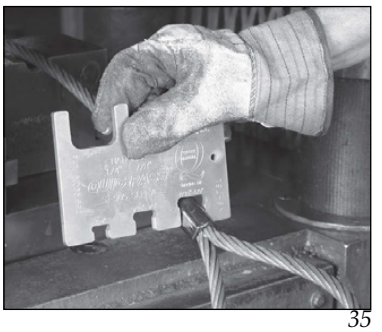
Dies must be fully closed in both passes before swaging is complete.



Step 7.
Open dies and check for proper after swage dimensions.

Use a National QUIC-PASS® No-Go Gauge to check the after swage dimensions to ensure that it has been swaged to the proper dimension. When swaged properly, the gauge will slide up and down the full length of the sleeve on all three sets of opposing flats. See page 18 for proper use of No-Go Gauges.





Note: Use only No-Go Gauges designed for QUIC-PASS® systems.

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Swage Fittings - Sockets

Swaging Methods for S-501 and S-502

National S-501 and S-502 Carbon Steel Swage Sockets are designed for swaging in National dies. These swage fittings are designed to be used on one (1) part of 6 X 19 or 6 X 37, IPS or XIP (EIP), RRL, FC, or IWRC wire rope. These products are designed to be used with inch size wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Cold Swaging Process

Cold swaging is a process which causes considerable movement and work hardening of the steel in the fitting. Exerting extreme pressures, the swaging machine forces the steel to flow into the crevices between the wires and strands of the wire rope. At the same time, the swaging machine forces the material to flow plastically along the length of the wire rope. These are a few of the reasons swaging is done in multiple passes in a well lubricated die.

Protecting Dies and Swaging Machine

When the die lands touch each other, effective swaging stops. Additional swaging adds needless wear and strain on the dies and swaging machine. We recommend placing oil on the die faces and cavities. When swaging, the oozing of the oil from the die faces will indicate when the dies are closed. At this point, stop swaging.

Avoid Excess Flashing

Flashing occurs when socket material flows out into the area between the die faces. Once formed, additional passes will not eliminate flashing. They will only fold back the flash and develop a permanent mark in the material and possibly create a crack.

To avoid excess flashing, do the following:

- Select proper die.
- Make sure that dies are in good condition and properly lubricated.
- Apply lubricant such as a light hydraulic oil to both die cavities and the lower die face.



CORRECT **EXCESS FLASHING**

- Follow swaging instructions on pages 39-41.



Swage sockets incorporate a reduced machined area of the shank which is equivalent to the proper after swage dimension. Before swaging, this provides for an obvious visual difference in the shank diameter. After swaging, uniform shank diameter is created allowing for a QUIC-CHECK® and permanent visual inspection opportunity.

Designed to quickly determine whether the socket has been through the swaging operation and assist in field inspections, it does not eliminate the need to perform standard production inspections which include gauging for the proper after swage dimensions or proof loading.

QUIC-CHECK®



U.S. Patent 5,152,630 and foreign equivalents.

Identification Stamping

For Identification purposes, National Swage products can be impression stamped after swaging. Properly done, there should be no concern for fractures initiating from such impressions.

To stamp S-501 and S-502 sockets:

- Use round corner or low stress stamps. Stamp to maximum depth of 0.015 in. (1/64”).
- Stamp in the area on the side of the socket head, as shown in the illustration.



NOTE: DO NOT STAMP ON SWAGING SHANK.

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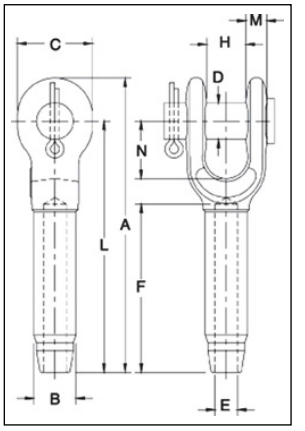
National Open Swage Sockets

QUIC-CHECK®

S-501



- Forged from special bar quality carbon steel, suitable for cold forming.
- Swage socket terminations have an efficiency rating of 100% based on the catalog strength of wire rope.
- Hardness controlled by spheroidize annealing.
- Stamp for identification after swaging without concern for fractures. (See page 36)
- Swage sockets incorporate a reduced machined area of the shank which is equivalent to the proper "After Swage" dimension. Before swaging, this provides for an obvious visual difference in the shank diameter. After swaging, a uniform shank diameter is created allowing for a **QUIC-CHECK®** and permanent visual inspection opportunity.
  - Designed to quickly determine whether the socket has been through the swaging operation and assist in field inspections, it does not eliminate the need to perform standard production inspections which include gauging for the proper "After Swage" dimensions or proof loading.
- U.S. Patent 5,152,630 and foreign equivalents.



NOTE: S-501 Swage Sockets are recommended for use with 6 x 19 or 6 x 37, IPS or XIP (EIP), XXIP (EEIP), RRL, FC or IWRC wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

In accordance with ASME B30.9, all slings terminated with swage sockets shall be proof loaded.*

S-501 Open Swage Sockets

S-501 and S-501B Open Swage Socket Specifications																Swager / Die Data			
		Rope Size*			Before Swage Dimensions (in.)												Stock No.		Side Load	
S-501 Stock No.	S-501B Stock No. **	(in.)	(mm)	Wt. Each (lbs.)	A	B	C	D	E	F	H	L	M	N	Max. After Swage Dim. (in.)	Die Description	500 1000 1500 Ton 5 x 7	1500 3000 Ton 6 x 12	1500 Ton 6 x 12	3000 Ton 6 x 12
1039021	1054001	1/4	6	.52	4.78	.50	1.38	.69	.27	2.19	.69	4.00	.38	1.47	.46	1/4 Socket	1192845	-	-	-
1039049	1054010	5/16	8	1.12	6.30	.78	1.62	.81	.34	3.25	.80	5.34	.48	1.67	.71	5/16-3/8 Socket	1192863	-	-	-
1039067	1054029	3/8	9-10	1.30	6.30	.78	1.62	.81	.41	3.25	.80	5.34	.48	1.67	.71	5/16-3/8 Socket	1192863	-	-	-
1039085	1054038	7/16	11-12	2.08	7.82	1.01	2.00	1.00	.49	4.31	1.00	6.69	.56	1.96	.91	7/16-1/2 Socket	1192881	-	-	-
1039101	1054047	1/2	13	2.08	7.82	1.01	2.00	1.00	.55	4.31	1.00	6.69	.56	1.96	.91	7/16-1/2 Socket	1192881	-	-	-
1039129	1054056	9/16	14	4.67	9.54	1.27	2.38	1.19	.61	5.38	1.25	8.13	.68	2.21	1.16	9/16-5/8 Socket	1192907	-	-	-
1039147	1054065	5/8	16	4.51	9.54	1.27	2.38	1.19	.68	5.38	1.25	8.13	.68	2.21	1.16	9/16-5/8 Socket	1192907	-	-	-
1039165	1054074	3/4	18-20	7.97	11.61	1.56	2.75	1.38	.80	6.44	1.50	10.00	.78	2.69	1.42	3/4 Socket	1192925	-	-	-
1039183	1054083	7/8	22	11.52	13.37	1.72	3.13	1.63	.94	7.50	1.75	11.63	.94	3.20	1.55	7/8 Socket	1192943	-	-	-
1039209	1054092	1	24-26	17.80	15.47	2.00	3.69	2.00	1.07	8.63	2.00	13.38	1.07	3.68	1.80	1 Socket	1192961	-	-	-
1039227	1054104	1-1/8	28	25.25	17.35	2.25	4.06	2.25	1.19	9.75	2.25	15.00	1.19	4.18	2.05	1-1/8 Socket	1192989	-	-	-
1039245	1054113	1-1/4	32	35.56	19.20	2.53	4.50	2.50	1.34	10.81	2.50	16.50	1.27	4.64	2.30	1-1/4 Socket	1193005	-	-	-
1039263	1054122	1-3/8	34-36	43.75	21.10	2.81	5.25	2.50	1.46	11.88	2.41	18.13	1.46	5.25	2.56	1-3/8 Socket	1193023	-	-	-
1039281	1054131	1-1/2	38-40	58.50	23.17	3.08	5.50	2.75	1.59	12.81	3.00	19.75	1.65	5.70	2.81	1-1/2 Socket	1193041	1191267	1195355	1195192
1039307	1054140	1-3/4	44	88.75	26.70	3.40	6.25	3.50	1.87	15.06	3.50	23.00	2.07	6.67	3.06	1-3/4 Socket	1193069	1191276	1195367	1195209
1042767	1054159	2	48-52	146.2	31.15	3.94	7.80	3.75	2.12	17.06	4.00	26.75	1.81	8.19	3.56	2 Socket	1193087	1191294	1195379	1195218

* Maximum Proof Load shall not exceed 50% of XXIP rope catalog breaking strength.
** Assembly with bolt, nut and cotter pin.

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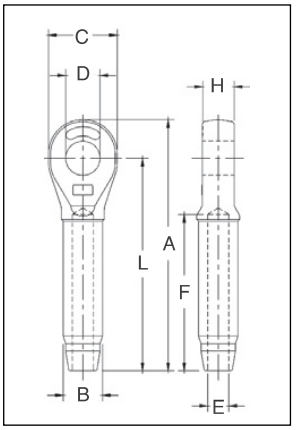
National Closed Swage Sockets

QUIC-CHECK®

S-502



- Forged from special bar quality carbon steel, suitable for cold forming.
- Swage socket terminations have an efficiency rating of 100% based on the catalog strength of wire rope.
- Hardness controlled by spheroidize annealing.
- Stamp for identification after swaging without concern for fractures. (See page 36)
- Swage sockets incorporate a reduced machined area of the shank which is equivalent to the proper "After Swage" dimension. Before swaging, this provides for an obvious visual difference in the shank diameter. After swaging, a uniform shank diameter is created allowing for a **QUIC-CHECK**® and permanent visual inspection opportunity.
  - Designed to quickly determine whether the socket has been through the swaging operation and assist in field inspections, it does not eliminate the need to perform standard production inspections which include gauging for the proper "After Swage" dimensions or proof loading.
- U.S. Patent 5,152,630 and foreign equivalents.



NOTE: S-502 Swage Sockets are recommended for use with 6 x 19 or 6 x 36, IPS or XIP (EIP), XXIP (EEIP), RRL, FC or IWRC wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

In accordance with ASME B30.9, all slings terminated with swage sockets shall be proof loaded.*

S-502 Closed Swage Sockets

S-502 Closed Swage Socket Specifications													Swage / Die Data				
	Rope Size*			Before Swage Dimensions (in.)								Max. After Swage Dim. (in.)		Stock No.		Side Load	
S-502 Stock No.	(in.)	(mm)	Wt. Each (lbs.)	A	B	C	D	E	F	H	L		Die Description	500 1000 1500 Ton 5 x 7	1500 3000 Ton 6 x 12	1500 Ton 6 x 12	3000 Ton 6 x 12
1039325	1/4	6	.33	4.28	.50	1.38	.76	.27	2.19	.50	3.50	.46	1/4 Socket	1192845	-	-	-
1039343	5/16	8	.75	5.42	.77	1.62	.88	.34	3.25	.68	4.50	.71	5/16-3/8 Socket	1192863	-	-	-
1039361	3/8	9-10	.72	5.42	.78	1.62	.88	.41	3.25	.68	4.50	.71	5/16-3/8 Socket	1192863	-	-	-
1039389	7/16	11-12	1.42	6.88	1.01	2.00	1.07	.49	4.31	.87	5.75	.91	7/16-1/2 Socket	1192881	-	-	-
1039405	1/2	13	1.42	6.88	1.01	2.00	1.07	.55	4.31	.87	5.75	.91	7/16-1/2 Socket	1192881	-	-	-
1039423	9/16	14	2.92	8.59	1.27	2.38	1.28	.61	5.38	1.14	7.25	1.16	9/16-5/8 Socket	1192907	-	-	-
1039441	5/8	16	2.85	8.59	1.27	2.38	1.28	.68	5.38	1.14	7.25	1.16	9/16-5/8 Socket	1192907	-	-	-
1039469	3/4	18-20	5.00	10.25	1.56	2.88	1.49	.80	6.44	1.33	8.63	1.42	3/4 Socket	1192925	-	-	-
1039487	7/8	22	6.80	11.87	1.72	3.12	1.73	.94	7.50	1.53	10.09	1.55	7/8 Socket	1192943	-	-	-
1039502	1	24-26	10.40	13.56	2.00	3.62	2.11	1.07	8.63	1.78	11.50	1.80	1 Socket	1192961	-	-	-
1039520	1-1/8	28	14.82	15.03	2.25	4.00	2.37	1.19	9.75	2.03	12.75	2.05	1-1/8 Socket	1192989	-	-	-
1039548	1-1/4	32	21.57	16.94	2.53	4.50	2.62	1.34	10.81	2.25	14.38	2.30	1-1/4 Socket	1193005	-	-	-
1039566	1-3/8	34-36	28.54	18.59	2.81	5.00	2.62	1.46	11.88	2.29	15.75	2.56	1-3/8 Socket	1193023	-	-	-
1039584	1-1/2	38-40	38.06	20.13	3.08	5.38	2.87	1.59	12.81	2.56	17.00	2.81	1-1/2 Socket	1193041	1191267	1195355	1195192
1039600	1-3/4	44	51.00	23.56	3.40	6.25	3.63	1.87	15.06	3.08	20.00	3.06	1-3/4 Socket	1193069	1191276	1195367	1195209
1042589	2	48-52	89.25	27.13	3.94	7.25	3.88	2.12	17.06	3.31	23.00	3.56	2 Socket	1193087	1191294	1195379	1195218

* Maximum Proof Load shall not exceed 50% of XXIP rope catalog breaking strength.

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Swage Socket Terminations – ¼" - 1"

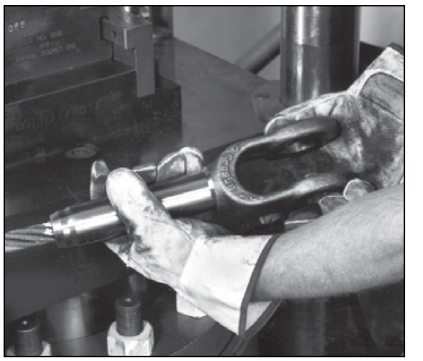
S-501 and S-502 Swage Sockets 1/4" through 1" (See page 40-41 for larger sizes).

S-501 and S-502 Swage Sockets are swaged in specially designed National Swage dies with a relief and extra length. **Small size sockets, through 1" should be swaged full shank.** To swage the socket, use the following procedure.

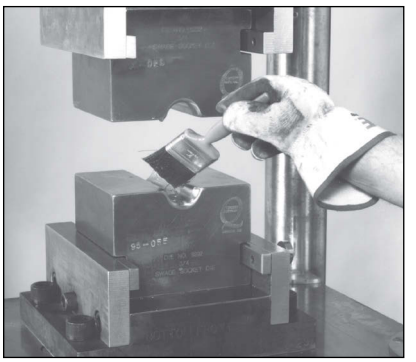
Step 1: For proper insertion, mark rope using Column F Dimension on page 37-38.



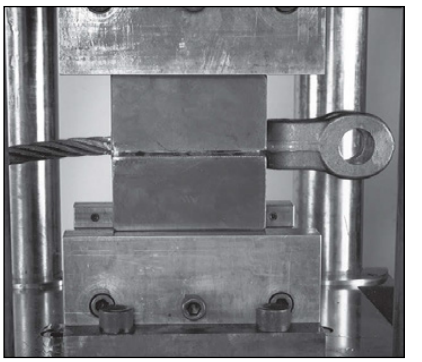
Step 2: Slide rope into socket until it is fully inserted in the shank. Inspect mark for full insertion.



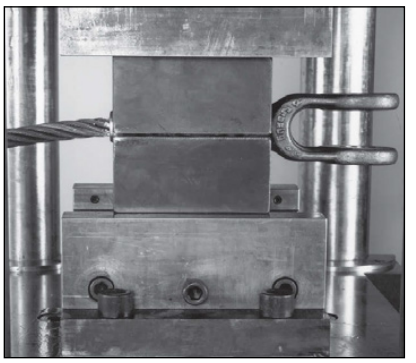
Step 3: Lubricate both die cavities and lower die face. See pages 37 and 38 for proper die selection.



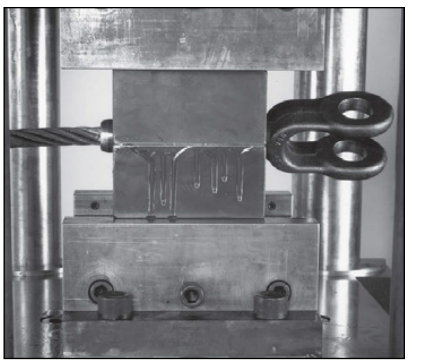
Step 4: 1st pass – allow dies to remain open approximately 1/2 the distance from time initial contact is made between socket and dies. Rotate fitting 45 to 90 degrees.



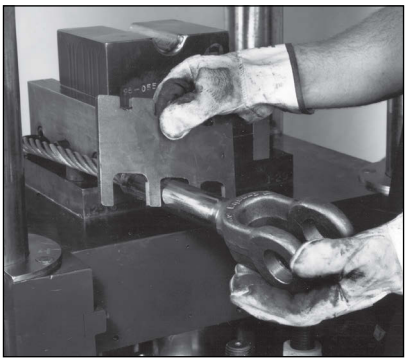
Step 5: 2nd pass – allow dies to remain open approximately ½ the distance from time initial contact is made between socket and dies. Rotate fitting 45 to 90 degrees.



Step 6: 3rd pass – If sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees.



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Step 7: 4th and 5th pass – Close dies to round fitting. Open dies and rotate 45 to 90 degrees between passes. **Open dies and check termination for proper after swage dimensions.** (See page 17 for proper use of No-Go Gauges).

Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed before swaging is complete.

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Swage Socket Terminations – 1⅛" & Larger

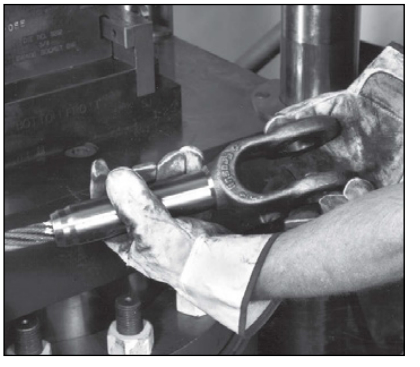
S-501 and S-502 Swage Sockets 1-1/8" and larger (See page 39 for smaller sizes).

S-501 and S-502 swage sockets are swaged in specially designed National Swage socket dies with a relief and extra length. **Sockets 1-1/8" and larger should be swaged full shank.** To swage the socket, use the following procedure.

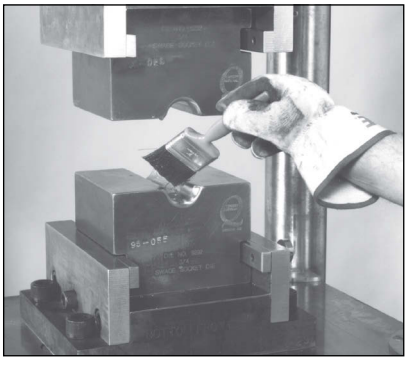
Step 1: For proper insertion, mark rope using Column F Dimension on page 37 or 38.



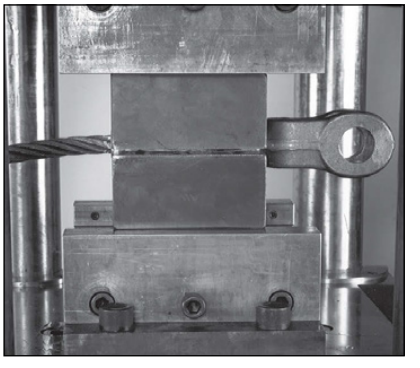
Step 2: Slide rope into socket until it is fully inserted in the shank. Inspect mark for full insertion.



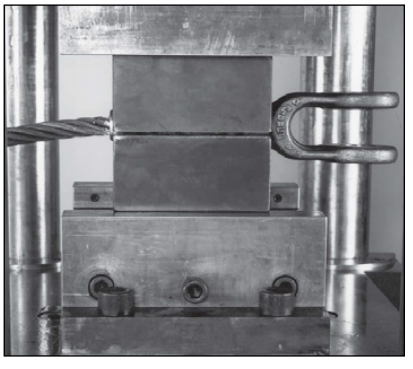
Step 3: Lubricate both die cavities and lower die face. See pages 37 and 38 for proper die selection.



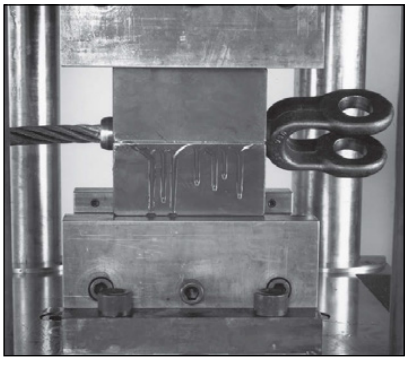
Step 4: 1st pass – allow dies to remain open approximately 1/2 the distance from time initial contact is made between socket and dies. Rotate fitting 45 to 90 degrees.

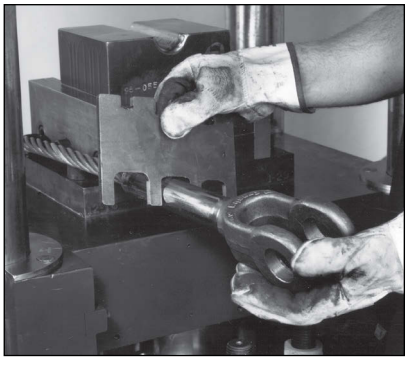


Step 5: 2nd pass – allow dies to remain open approximately 1/2 the distance from time initial contact is made between socket and dies. Rotate fitting 45 to 90 degrees.



Step 6: 3rd pass – If sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees.





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Step 7: 4th pass – Close dies to round fitting. **Open dies and check for proper after swage dimensions.** (See page 17 for proper use of No-Go Gauges).

Note: Limited swaging capacity may require the use of progressive swaging (See page 41).

Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed before swaging is complete.

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Swage Socket Terminations – Progressive Swaging

S-501 and S-502 swage sockets, and other fittings, must be swaged progressively if one of the following conditions exist.

1. The die Working Load Limit (WLL) is lower than the required swaging tonnage. **Warning – you must adjust tonnage so maximum die Working Load Limit is not exceeded. Failure to do this can result in serious injury or death.**

2. The Swaging Machine Capacity is lower than the required tonnage to swage the socket full shank.

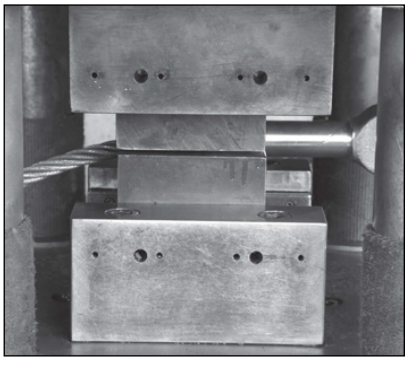
Before swaging make sure you follow all instructions in this manual.

NOTE: For additional information, see Swaging Machine Capacity Chart on page 12.

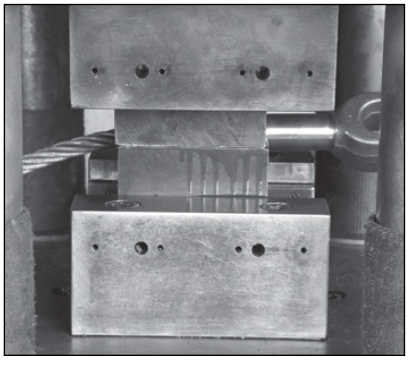
Step 1. Lubricate both die cavities and lower die face. See pages 37 and 38 for proper die selection.



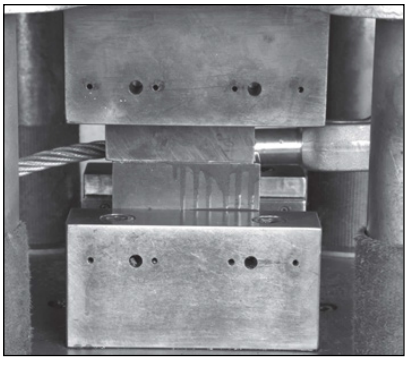
Step 2. Insert tapered end, or rope end of socket, about one third (1/3) of the shank into the proper die. Close the die until it comes in contact with the shank, and continue closing dies to approximately one-half (½) the distance between the dies.



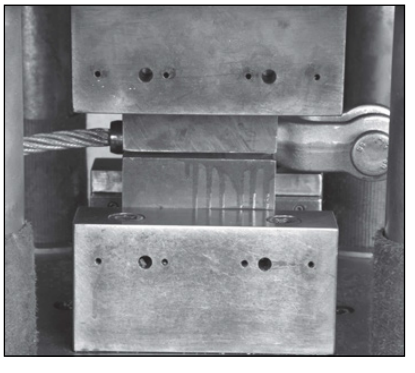
Step 3. Open dies and lubricate the socket shank. Rotate the socket shank 45 to 90 degrees and repeat closing the die to about one-half (½) the distance from the previous pass.



Step 4. Open dies and lubricate the socket shank. Rotate the socket shank 45 to 90 degrees and close the die.



Step 5. Open dies and insert the next one-third (1/3) into the die and repeat the process. This should be done until the full shank is swaged. Additional passes may be required to round out the shank and insure the correct after swage dimensions are met.



SEE APPLICATION AND WARNING INFORMATION
On Pages 11-18

Failure to reduce Swaging Machine tonnage to Working Load Limit (WLL) of die can cause serious injury or death. For instructions on adjusting tonnage see Die Working Load Limit Pressure Adjustment section of Swaging Machine Application Instructions (See Page 12).

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COLD TUFF® Swage Fittings – Buttons

Swaging Methods for S-409 Buttons

National S-409 Steel Buttons are designed for use with swaging in National dies. These swage fittings are designed to be used on one (1) part of 6 X 19 or 6 X 37 FC IPS or IWRC EIPS Right Regular Lay wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.



Cold Swaging Process

Cold swaging is a process which causes considerable movement and work hardening of the steel in the fitting. Exerting extreme pressures, the swaging machine forces the steel to flow into the crevices between the wires and strands of the wire rope. At the same time, the swaging machine forces the material to flow plastically along the length of the wire rope. These are a few of the reasons swaging is done on steel fittings in multiple passes in a well lubricated die.

Protecting Dies and Swaging Machines

When the die lands touch each other, effective swaging stops. Additional swaging adds needless wear and strain on the dies and swaging machine. We recommend placing oil on the die faces and cavities. When swaging, the oozing of the oil from the die faces will indicate when the dies are closed. At this point, stop swaging.

Avoid Excess Flashing

Flashing occurs when sleeve material flows out into the area between the die faces. Once formed, additional passes will not eliminate flashing. They will only fold back the flash and develop a permanent mark in the material and possibly create a crack.

To avoid excess flashing, do the following:

- Select proper dies.
- Make sure that dies are in good condition and properly lubricated.
- Apply lubricant such as a light hydraulic oil to both die cavities and the lower die face.
- Follow swaging instructions on pages 44-45.



CORRECT **EXCESS FLASHING**

Identification Stamping

For identification purposes, COLD TUFF® swage products can be impression stamped after swaging. Properly done, there should be no concern for fractures initiating from such impressions.



To stamp National S-409 Buttons

- Use round corner or low stress stamps. Stamp to maximum depth of 0.015 in. (1/64”).
- Stamp in the area on the side of the button, in the plane of the rope, no less than 0.250 in. (1/4”) from either end of the sleeve.

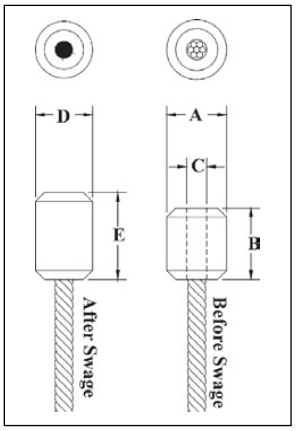
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National Swage Buttons

S-409



- Swage Button terminations have an efficiency rating of 98% based on the catalog strength of wire rope.
- Special processed, low carbon steel.
- COLD TUFF® for better swageability.
- Stamp for identification after swaging without concern for fractures. (See page 42)



NOTE: S-409 Buttons are recommended for use with 6 x 19 or 6 x 37, IPS or XIP (EIP), RRL, FC or IWRC wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

S-409 COLD TUFF® Buttons

S-409 Steel Swage Button Specifications											Swager / Die Data	
		Rope Size			Before Swage Dimensions (in.)			After Swage Dimensions (in.)				
S-409 Stock No.	Size No.	(in.)	(mm)	Weight Per 100 Pcs. (lbs.)	A	B	C	D Maximum After Swage Dimensions	E Length*		Die Description	Stock No. 500 Tons 1000 Tons 1500 Tons 5 x 7
1040171	1 SB	1/8	3	2	.42	.50	.14	.40	.61		1/8 - 1/4 Button	1191621
1040215	3 SB	3/16	5	4	.56	.70	.20	.52	.84		1/4 1st Stage	1197528
1040251	5 SB	1/4	6-7	8	.68	1.06	.31	.58	1.41		1/8 - 1/4 Button	1191621
1040297	7 SB	5/16	8	16	.88	1.13	.36	.77	1.33		3/8 1st Stage	1192364
1040313	8 SB	3/8	9-10	15	.88	1.48	.42	.77	1.69		3/8 1st stage	1192364
1040331	9 SB	7/16	11	30	1.13	1.63	.48	1.03	1.94		1/2 1st Stage	1192408
1040359	10 SB	1/2	13	50	1.31	1.89	.55	1.16	2.17		5/8 Socket	1192907
1040377	11 SB	9/16	14	70	1.44	2.02	.61	1.29	2.41		9/16 -5/8 Button	1191665
1040395	12 SB	5/8	16	100	1.56	2.42	.67	1.42	2.89		3/4 Socket	1192925
1040411	13 SB	3/4	18-20	131	1.68	2.74	.80	1.55	3.25		3/4 1st Stage	1192462
1040439	14 SB	7/8	22	220	2.00	3.27	.94	1.80	3.86		7/8 1st Stage	1192480
1040457	15 SB	1	25-26	310	2.25	3.67	1.06	2.05	4.36		1 1st Stage	1192505
1040475	16 SB	1-1/8	28-29	450	2.56	4.05	1.19	2.30	4.81		1-1/8 1st Stage	1192523
1040493	17 SB	1-1/4	31-32	650	2.81	4.57	1.33	2.56	5.42		1-3/8 Socket	1193023

* NOTE: Length is measured from outside end of termination.

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Button Terminations – 1/8" - 7/8"

S-409 Button Terminations 1/8" (#1) through 7/8" (#14) (See page 45 for larger sizes).

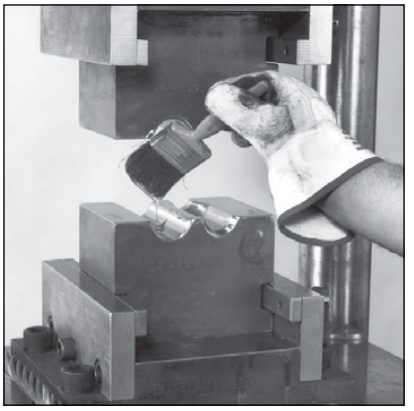
Buttons are swaged in open channel dies. Refer to the chart on page 43 for proper die selection.

To swage the button, use the following procedure:

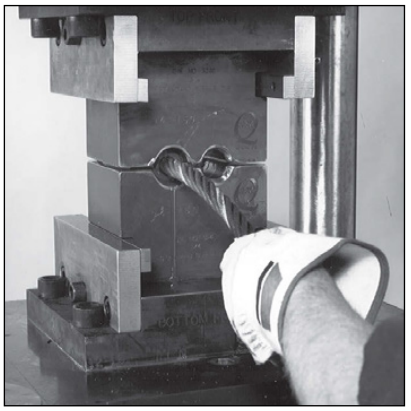
Step 1: Slide wire rope through button until the end of the wire rope protrudes a minimum of 1/8" (3mm) rope diameter from the end of the button. After swaging, the rope should be about flush with the button end.



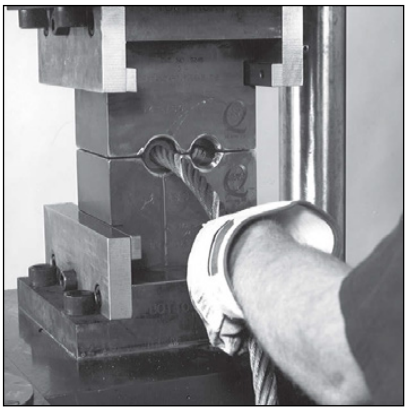
Step 2: Lubricate both die cavities and lower die face. (See page 43 for proper die selection).



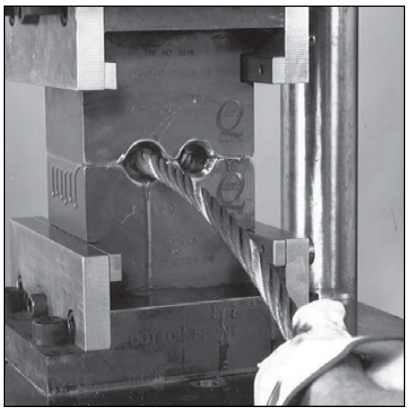
Step 3: 1st pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the button and dies. Open dies and rotate fitting 45 to 90 degrees.



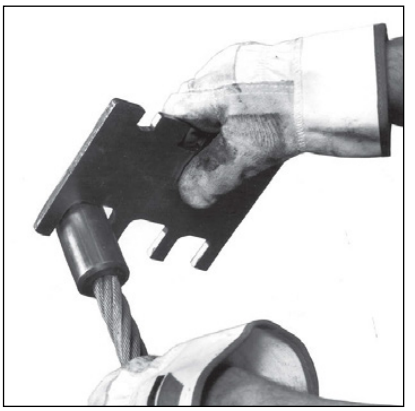
Step 4: 2nd and 3rd passes – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the button and dies. Open dies and rotate fitting 45 to 90 degrees between passes.



Step 5: 4th pass – if sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees.



Step 6: 5th and 6th passes – Rotate fitting 45 degrees (1/8 turn). Close dies to round fitting. Open dies and rotate fitting 45 to 90 degrees between passes. **Open dies and check termination for proper after swage dimensions.** (See page 17 for proper use of No-Go Gauges).



Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed before swaging is complete.

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Button Terminations – 1″ & Larger

S-409 Button Terminations 1” (#15) through 1-1/4” (#17) (See page 44 for smaller sizes).

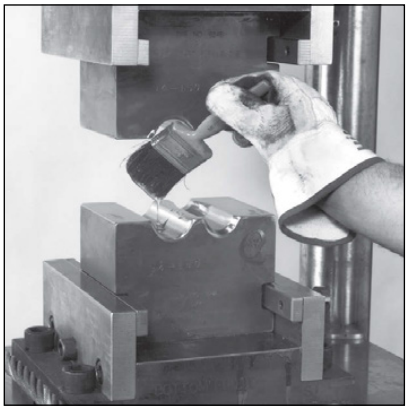
Buttons are swaged in open channel dies. Refer to chart on page 43 for proper die selection.

To swage the button, use the following procedure:

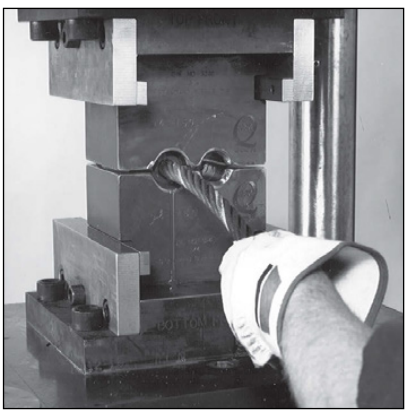
Step 1: Slide wire rope through button until the end of the wire rope protrudes a minimum of 1/8” (3mm) from the end of the button. After swaging, the rope should be about flush with the button end.



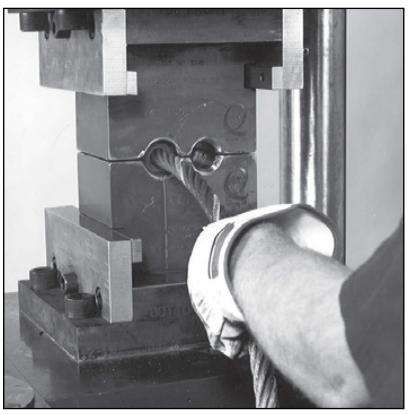
Step 2: Lubricate both cavities and lower die face. See page 43 for proper die selection.



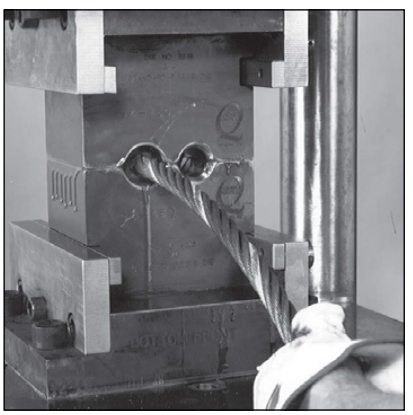
Step 3: 1st pass – allow dies to remain open approximately ½ the distance from the time initial contact is made between the button and dies. Open dies and rotate fitting 45 to 90 degrees.



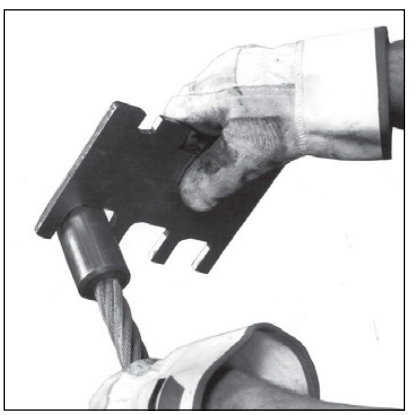
Step 4: 2nd, 3rd and 4th passes – allow dies to remain open approximately ½ the distance from the time initial contact is made between the button and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 5: 5th pass – if sharp flashing does not occur, close dies. Open dies and rotate fittings 45 to 90 degrees.



Step 6: 6th and 7th passes – Close dies to round fitting. Open dies and rotate fitting 45 to 90 degrees between passes. **Open dies and check termination for proper after swage dimensions.** (See page 17 for proper use of No-Go Gauge).



Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed before swaging is complete.

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COLD TUFF® Swage Fittings – Duplex Sleeves

Swaging Methods for National S-506 Turnback Eyes

National S-506 carbon steel sleeves are designed for use with turnback eye splices and for swaging in National dies.

These swage fittings are designed to be used on one (1) part of 6 X 19 or 6 X 37 FC IPS or IWRC EIPS Right Regular Lay wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Cold Swaging Process

Cold swaging is a process which causes considerable movement and work hardening of the steel in the fitting. Exerting extreme pressures, the swaging machine forces the steel to flow into the crevices between the wires and strands of the wire rope. At the same time, the swaging machine forces the material to flow plastically along the length of the wire rope. These are a few of the reasons swaging is done on steel fittings in multiple passes in a well lubricated die.

Protecting Dies and Swaging Machines

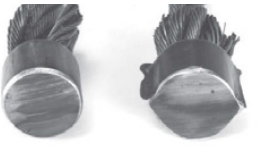
When the die lands touch each other, effective swaging stops. Additional swaging will add needless wear and strain on the dies and swaging machine. We recommend placing oil on the die faces and cavities. When swaging, the oozing of the oil from the die faces will indicate when the dies are closed. At this point, stop swaging.

Avoid Excess Flashing

Flashing occurs when sleeve material flows out into the area between the die faces. Once formed, additional passes will not eliminate flashing. They will only fold back the flash and develop a permanent mark in the material and possibly create a crack.

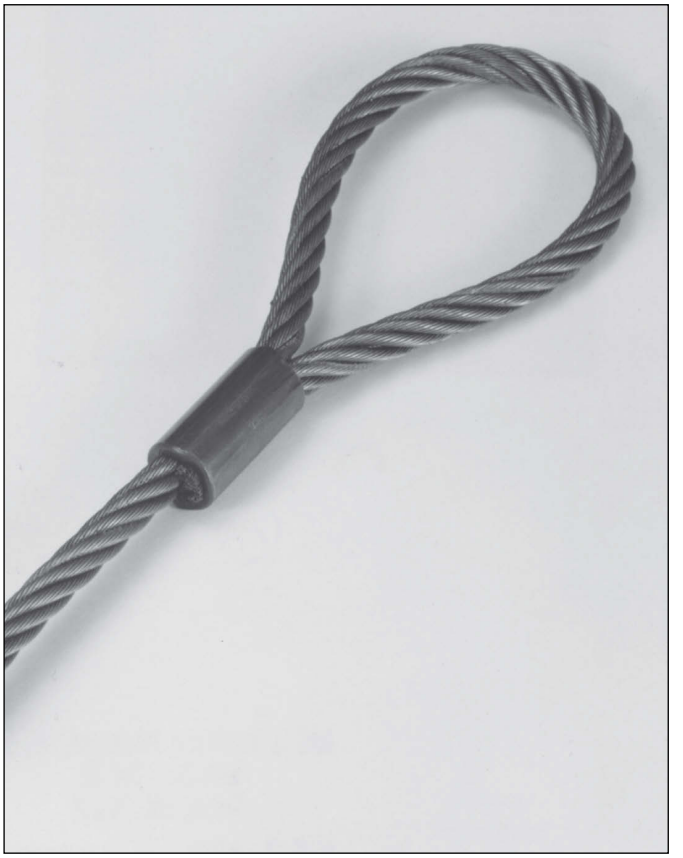
To avoid excess flashing, do the following:

- Select proper dies.
- Make sure that dies are in good condition and properly lubricated.
- Apply lubricant such as a light hydraulic oil to both die cavities and the lower die face.



CORRECT **EXCESS FLASHING**

- Follow swaging instructions on pages 48-49.



Identification Stamping

For identification purposes, National COLD TUFF® swage products can be impression stamped after swaging. Properly done, there should be no concern for fractures initiating from such impressions.



To stamp S-506 sleeves:

- Use round corner or low stress stamps. Stamp to maximum depth of 0.015 in. (1/64”).
- Stamp in area on the side of the sleeve, in the plane of the sling eye, and no less than 0.250 in. (1/4”) from either end of the sleeve.

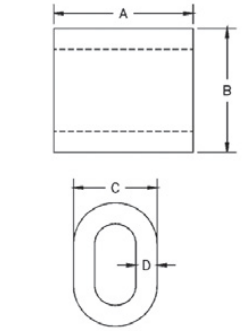
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National Swage Duplex Sleeves

S-506



- For turnback wire rope splicing.
- Special processed low carbon steel.
- Turnback terminations have efficiency ratings of 94% based on the catalog strength of wire rope.
- Designed for lower temperature toughness.
- Resists cracking when swaged (equals or exceeds stainless steel sleeves).
- COLD TUFF® for better swageability.
- Stamp for identification after swaging without concern for fractures. (See page 46)



NOTE: S-506 Duplex sleeves are recommended for use with 6 x 19 or 6 x 37, IPS or XIP (EIP), RRL, FC or IWRC wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

S-506 COLD TUFF® Duplex Non-Tapered Sleeves

S-506 Steel Duplex Non-Tapered Sleeve Specifications										Swager / Die Data	
S-506 Stock No.	Rope Size (in.)	Rope Size (mm)	Weight Per 100 Pcs. (lbs.)	Pkg. Qty.	Before Swage Dimensions (in.) A	B	C	D	Max. After Swage Dimensions (in.)	Die Description	Stock No. 500 Tons 1000 Tons 1500 Tons 5 x 7
1039334	5/16	8	17	200	1.25	1.06	.81	.19	.77	3/8 1st Stage	1192364
1039352	3/8	9-10	13	100	1.25	1.12	.81	.14	.77	3/8 1st Stage	1192364
1039370	7/16	11	31	50	1.63	1.41	1.02	.19	1.03	1/2 1st Stage	1192408
1039398	1/2	13	27	50	1.63	1.44	1.02	.16	1.03	1/2 1st Stage	1192408
1039414	9/16	14	63	25	2.25	1.72	1.23	.23	1.29	5/8 1st Stage	1192444
1039432	5/8	16	54	25	2.25	1.84	1.28	.20	1.29	5/8 1st Stage	1192444
1039450	3/4	18-20	91	10	2.63	2.16	1.52	.23	1.55	3/4 1st Stage	1192462
1039478	7/8	22	126	10	2.88	2.50	1.75	.27	1.80	7/8 1st Stage	1192480
1039496	1	25-26	187	10	3.06	2.84	2.00	.33	2.05	1 1st Stage	1192505
1039539	1-1/4	30-32	384	Bulk	4.06	3.50	2.50	.38	2.56	1-3/8 Socket	1193023

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Turnback Terminations – 5/16" - 5/8"

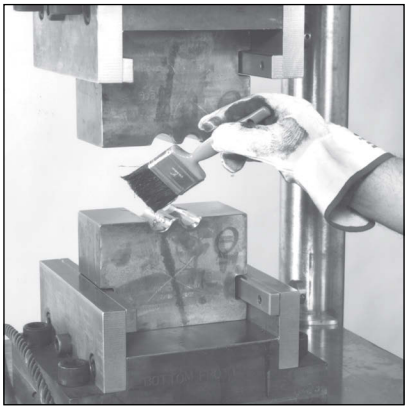
Turnback Eye Terminations using 5/16" through 5/8" S-506 sleeves utilize the open cavity of the S-505 Standard Steel Sleeve die on a National Hydraulic Swaging Machine. (See page 49 for larger sizes.) Each S-505 Standard Steel Sleeve die contains 2 cavities (open and tapered).

To form the eye and swage the termination, use the following procedure:

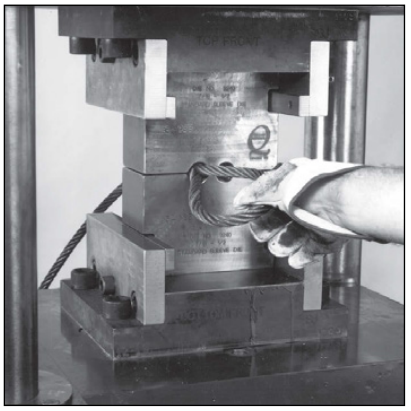
Step 1: Slide the end of the rope through the sleeve and turn back the wire rope end through the S-506 Duplex Steel Sleeve and adjust the rope to form the proper eye size. Allow approximately 1 (one) rope diameter to extend past the end of the sleeve. This will provide for elongation of the fitting during swaging. **See page 47 for proper die selection.**



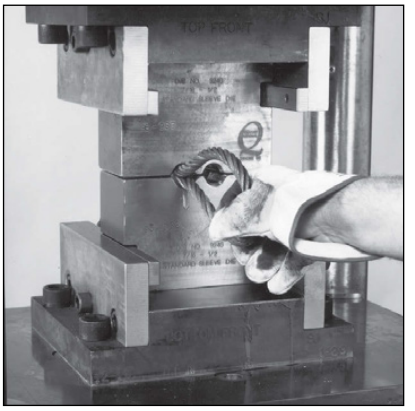
Step 2: Lubricate both die cavities and lower die face.



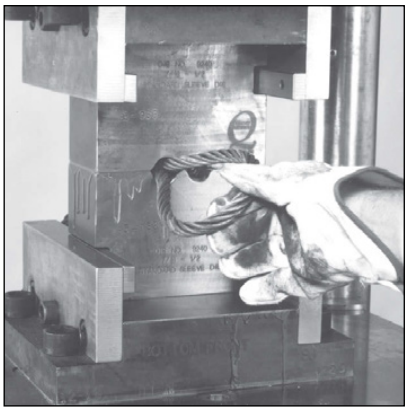
Step 3. 1st pass – allow dies to remain open approximately 1/2 the distance from time initial contact is made between sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



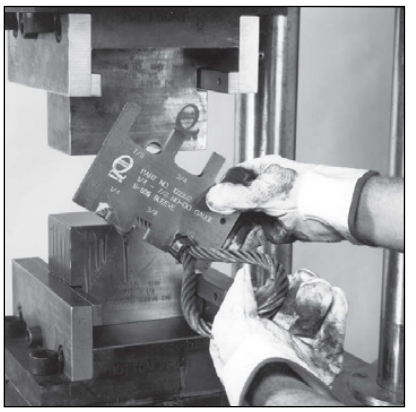
Step 4. 2nd pass – allow dies to remain open approximately 1/2 the distance from the time initial contact is made between the sleeve and dies. Open dies and rotate fitting 45 to 90 degrees.



Step 5: 3rd pass – If sharp flashing does not occur, close dies. Open dies and rotate fitting 45 to 90 degrees.



Step 6: 4th and 5th pass – Close dies to round fitting. Open dies and rotate fitting 45 to 90 degrees between passes. **Open dies and check for proper after swage dimension.** (See page 17 for proper use of No-Go Gauge).



Additional swaging may be required to provide smooth finish and achieve proper after swage dimensions.
Dies must be fully closed before swaging is complete.

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